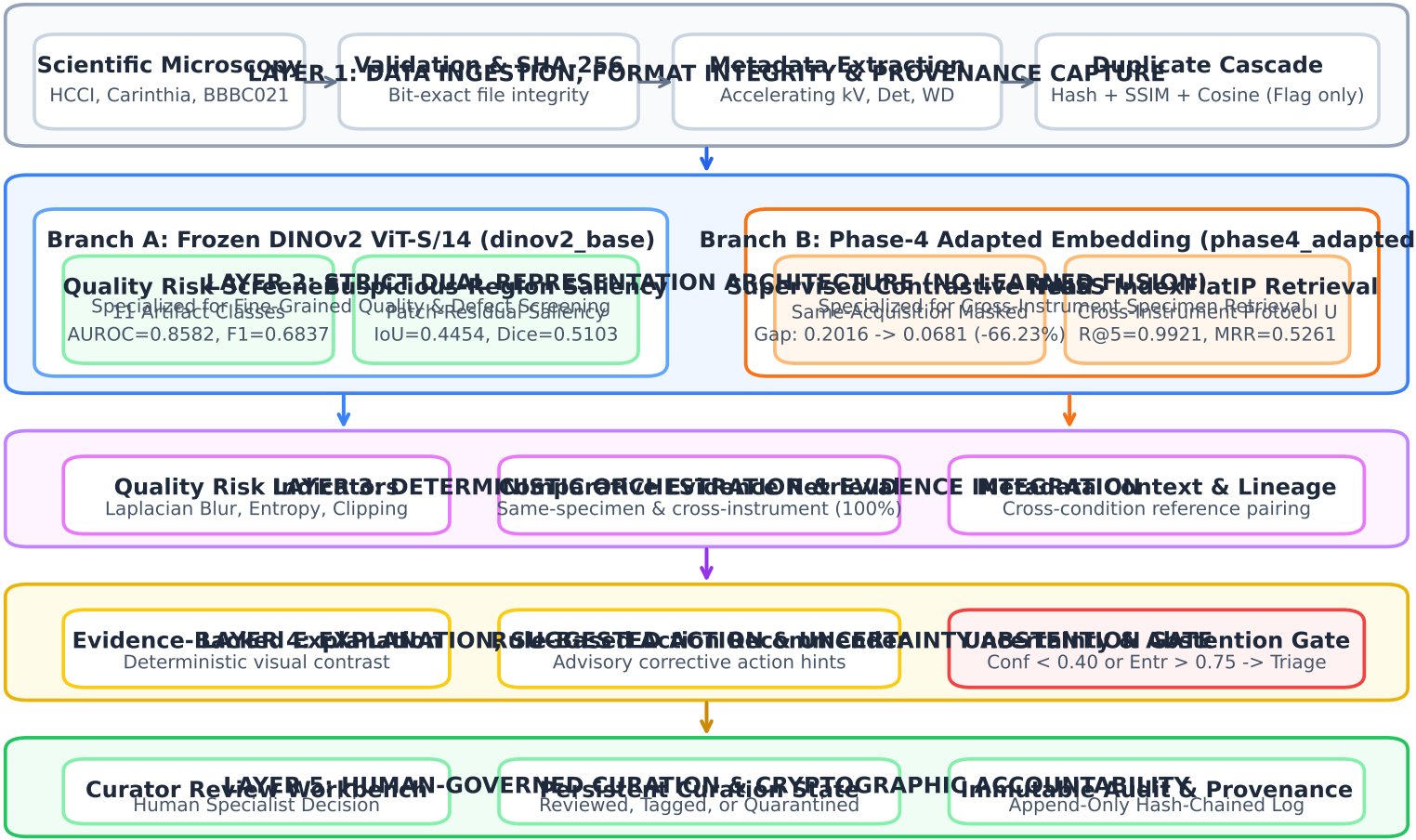
LAYER 1: DATA INGESTION, FORMAT INTEGRITY & PROVENANCE CAPTURE
Scientific Microscopy
HCCI, Carinthia, BBBC021
Validation & SHA-256
Bit-exact file integrity
Metadata Extraction
Accelerating kV, Det, WD
Duplicate Cascade
Hash + SSIM + Cosine (Flag only)
LAYER 2: STRICT DUAL-REPRESENTATION ARCHITECTURE (NO LEARNED FUSION)
Branch A: Frozen DINOv2 ViT-S/14 (dinov2_base)
Quality Risk Screener
Specialized for Fine-Grained Quality & Defect Screening
11 Artifact Classes
AUROC=0.8582, F1=0.6837
Suspicious Region Saliency
Patch-Residual Saliency
IoU=0.4454, Dice=0.5103
Branch B: Phase-4 Adapted Embedding (phase4_adapted)
Supervised Contrastive Heads
Specialized for Cross-Instrument Specimen Retrieval
Same-Acquisition Masked
Gap: 0.2016 -> 0.0681 (-66.23%)
LatIP Retrieval
Cross-Instrument Protocol U
R@5=0.9921, MRR=0.5261
LAYER 3: DETERMINISTIC ORCHESTRATION & EVIDENCE INTEGRATION
Quality Risk Labelers
Laplacian Blur, Entropy, Clipping
Cross-Instrument Retrieval
Same-specimen & cross-instrument (100%)
Metadata Context & Lineage
Cross-condition reference pairing
LAYER 4: EXPLANATION, SUGGESTED ACTION & UNCERTAINTY ABSTENTION GATE
Evidence-Backed Explanations
Deterministic visual contrast
Suggested Actions
Advisory corrective action hints
Uncertainty Abstention Gate
Conf < 0.40 or Entr > 0.75 -> Triage
LAYER 5: HUMAN-GOVERNED CURATION & CRYPTOGRAPHIC ACCOUNTABILITY
Curator Review Workbench
Human Specialist Decision
Persistent Curation State
Reviewed, Tagged, or Quarantined
Immutable Audit & Provenance
Append-Only Hash-Chained Log